Econ 452: Final Assignment

Due Date: March 20th

Rules

This assignment focuses on the BLP model. The first seven questions ask you to apply the estimator to a new dataset, interpret the output, and conduct counterfactual simulations. These exercises closely follow the applications we covered in class.

Please submit a single PDF file containing your written answers, code, and output. Please attach the problem set questions (stated here) to the end of your submission. You may just add it to the end of the submission, this facilitates the review. You can either:

- Email the PDF to `yiyangchen@g.ucla.edu` , or
- Upload it directly to the course website.

The submission must be received by the date and time indicated above. You may work in groups of up to three students and submit one write-up per group. I strongly encourage you to collaborate.

Video Quiz (Question 8): After submitting the written assignment, each student must also complete a short individual video quiz. This will consist of one or two questions related to the BLP model and the assignment, and your response should be recorded in a 5-minute video. Each group member must answer the quiz independently and upload their recording to BruinLearn. We will share additional details about the video quiz in the coming days.

Data

Download the two datasets—`agents_pizza.csv` and `products_pizza.csv`—that you will use for this assignment. `agents_pizza.csv` contains simulated data on representative pizza consumers in Westwood, including a market identifier, random taste shocks on a constant term (node0) and price (node 1), more detail below, and the consumer's inverse income $invincome = \frac{1}{Income}$. `products_pizza.csv` contains simulated product and firm level information on pizzas in Westwood. Each row corresponds to a different product. The data include a market identifier, a firm identifier, the product's market share and four product characteristics, including the price (measured in $ per slice),the product's quality rating, the average delivery time to consumers (denoted in hours), and the hourly (labor) wage costs in $100. The data also include 5 demand and 6 supply instruments. When values about select variables are missing, think about a way to use the data to impute the values.

1 Environment

1.1 Demand

We study the competition between pizza bakers in Westwood that offer delivery services to consumers situated in Westwood. Consumer i indirect utility function for pizza j in market t is as follows:

$$
\begin{aligned}
u_{ijt} &= \beta_i^p \times p_{jt} + \beta^d \times deliverytime_{jt} + \beta^r \times rating_{jt} + \xi_{jt} + \epsilon_{ijt} && (1)\\
u_{i0t} &= \alpha_i + \epsilon_{i0t} \ . && (2)
\end{aligned}
$$

Here, and following the structure of the BLP model covered in class, p_{jt} denotes the price and ξ_{jt} denotes unobserved product quality. $deliverytime_{jt}$ and $rating_{jt}$ denote the observed product characteristics discussed above. We assume the ϵ_{ijt} are distributed i.i.d. extreme value and allow for preference heterogeneity over prices and a constant term for the outside good denoted by 0. Specifically, we assume that

$$
\begin{aligned}
\beta_i^p &= \beta^p + \pi \times \frac{1}{Income_i} && (3)\\
\alpha_i &\sim N(0, \sigma^2) && (4)
\end{aligned}
$$

where again similar to the BLP problem, α_i and $\frac{1}{Income_i}$ denote random coefficients.

1.2 Supply

We model supply using a standard multiproduct-firms differentiated-product Bertrand model. Pizza bakers can produce pizzas at marginal costs:

$$
mc_{jt} = \gamma_0 + \gamma_1 \times rating_{jt} + \gamma_3 \times hourlywages_{jt} + \gamma_4 \times deliverytime_{jt} + \omega_{jt} \qquad (5)
$$

Pizza bakers take their marginal costs and the exogenous demand shifters (rating and deliverytime) as given and choose the profit-maximizing prices.

2 Analysis

Question 1: Instrumental Variables

Building on the in-class discussion, we use demand and supply instruments to identify the structural model parameters.

1. Demand: Explain (in theory) why BLP instruments can be used to estimate demand parameters taking price endogeneity into account.

2. Demand: In this exercise, we use two sets of instruments:

   - Sums over characteristics of non-rival goods: $Z_{jt}^{BLP,other}(X) = \sum_{(k \in J_{ft} \setminus j)} x_{kt}$
   - Sums over characteristics of rival goods: $Z_{jt}^{BLP,rival}(X) = \sum_{(k \notin J_{ft})} x_{kt}$

   Building on these definitions, explain verbally how $demand_instruments0$-$demand_instruments3$ as well as $supply_instruments0$-$supply_instruments5$ in `products_pizza.csv` are exactly constructed.

3. What are shocks can be used as demand instruments and how to the relate to $demand_instruments4$?

Question 2: Model Estimation

Using the `pyblp package`, load the agent and the product data and estimate the model.

- Formulate the model, which requires specifying the linear demand parameters, the non-linear demand parameters, and the marginal cost function.
- Formulate the agent model (which specifies the variables involving interacted with random coefficients or patient demographics)
- Specify starting values and the dimensionality of the parameter space.
  - sigma is a $l \times l$ matrix, where l denotes the number of non-linear variables. (Note: sigma captures the variance covariance matrix of the random coefficients and since we do not have a random coefficient on price (only demographic shifter), you can set the corresponding diagonal and off-diagonal parameters to zero in the problem formulation)
  - π is a l vector, where l denotes the number of non-linear variables. (Note: π coefficients on the demographic shifters. Since we do not have a demographic shifter for the constant term, you can set the corresponding parameter to zero in the problem formulation)
  - β: please also specify starting values for the linear parameters (you may otherwise encounter some computational problems)

Discuss the parameter estimates:

- Explain in one sentence (each) the linear demand parameter (β) effect sizes.
- Explain in one sentence (each) the marginal cost parameter (γ) effect sizes.
- Explain in one sentence (each) the nonlinear coefficient estimates (sigma and π)

Question 3: Elasticities and Diversion Ratios

Take an example market and:

- Summarize the own and cross-price elasticities. You may use a graph for this or simply calculate an average across products. Discuss in one sentence how you interpret the own- and the cross-price elasticity
- Summarize the diversion ratios. Discuss in one sentence how you interpret the diversion ratios for inside and the outside good.

Question 4: Marginal Costs, Markups, HHI, Consumer Surplus, and Profits

1. Construct and summarize the distribution of marginal costs across products. Interpret the magnitude of the average marginal cost estimate.
2. Construct and summarize the distribution of markups, $markup = \frac{(p-mc)}{p}$, across products. Interpret the magnitude of the average markup estimate. Using the Lerner index, related the estimated average markup to the average price elasticity derived above.
3. Construct and summarize the distribution of the Herfindahl-Hirschman index (HHI) across markets. Is any of the markets highly concentrated (HHI$\geq$2,500)?

4. Construct and summarize the distribution of profits (across products) assuming a full pizza has 6 slices and there in total 100 thousand consumers in the market for pizzas. What's the average profit per product?

5. Construct and summarize the distribution of consumer surplus (across markets) a full pizza has 6 slices and there in total 100 thousand consumers in the market for pizzas. What's the average consumer surplus per market?

Question 5: Merger Analysis

Now suppose that Fresh Brothers (firm id 1) acquires Italian Express Pizza (firm id 0), Lamonica's NY Pizza (firm id 2), and Enzo's Pizzeria (firm id 3).

1. Construct the new equilibrium prices and market shares. Summarize the distribution of changes in the HHI across markets.

2. Construct and summarize the changes in markups across products. By how much markups increase or decrease on average and why are they increasing/decreasing?

3. Construct and summarize the changes in profits across products. By how much markups increase or decrease on average and why are they increasing/decreasing?

4. Construct and summarize the changes in consumer surplus across markets assuming there are in total 100 thousand consumers. By how much markups increase or decrease on average and why are they increasing/decreasing?

Question 6: Delivery Ceiling

Now suppose the acquisitions of Fresh Brothers had not occurred and we are back in the initial markets structure with 6 competing firms. Instead the major decides that pizzas must be delivered within 33 minutes. We do not consider how delivery drivers achieve the faster times but assume that all deliveries that would have taken longer than 33 minutes actually take only 33 minutes (which affects demand and costs).

1. Construct the new equilibrium prices and market shares at the estimated parameters (and holding unobserved demand (ξ) and cost shocks (ω) fixed. Summarize the changes in prices across products.

2. Construct the new marginal costs and summarize the changes in marginal costs across products. By how much markups increase or decrease on average and why are they increasing/decreasing?

3. Construct the new markups and summarize the changes in markups across products. Do you have some intuition on why markups may have increased/decreased on average?

4. Construct the new consumer surplus and summarize the changes in consumer surplus across markets assuming there are in total 100 thousand consumers. Do you have some intuition on why consumer surplus may have increased/decreased on average?

Question 7: Gentrification

Now suppose the acquisitions of Fresh Brothers had not occurred and there is no deliverytime regulation. Instead suppose that there is an influxe of higher income indivuals to the market. The number of indivuals with an inverse income of less than 0.2 increases 5-fold. They have the same distribution of unobserved random coefficients and income than their peers. Discuss briefly in your own words how you would go about simulating the changes in consumer demand.

1. Construct the new equilibrium prices and market shares at the estimated parameters (and holding unobserved demand (ξ) and cost shocks (ω) fixed. Summarize the changes in prices across products, why are they increasing/decreasing?

2. Construct the new marginal costs and summarize the changes in marginal costs across products. By how much markups increase or decrease on average and why are they increasing/decreasing?

3. Construct the new markups and summarize the changes in markups across products. Do you have some intuition on why markups may have increased/decreased on average?

Question 8: Video Quiz

In this last exercise, we ask you to verbally respond to one question related to the BLP model and your assignments. To this end, we ask you to video-record a 5 minute answer and upload the video to bruinlearn. More details on the logistics will follow soon.